Statistical Analysis of Team Strengths, Match Performance, and Comparator Modeling in the IPL (MI, RCB, CSK, KKR)

Project Analysis Report

2022–2026 Season Analysis

1 Introduction & Project Overview

This research project analyzes the performance of four prominent Indian Premier League (IPL) franchises: **Mumbai Indians (MI)**, **Royal Challengers Bengaluru (RCB)**, **Chennai Super Kings (CSK)**, and **Kolkata Knight Riders (KKR)**. Using granular ball-by-ball data from 2022 to 2026, we quantify team batting and bowling strengths based on multi-factor weighted metrics, evaluate match-by-match performance, and establish a predictive comparator framework incorporating home-ground advantage and opponent adjustments.

2 Methodology & Mathematical Formulation

2.1 Batting Strength Formulation

Batting strength is computed using normalized performance metrics mapped between 0 and 1. For higher-is-better metrics ($x_{\max}$ to $x_{\min}$):

$$N(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

For lower-is-better metrics (such as dot-ball percentage):

$$N(x) = \frac{x_{\max} - x}{x_{\max} - x_{\min}} \tag{2}$$

The final composite **Batting Strength** is calculated as:

$$\text{BattingStrength} = 0.30N(\text{Runs}) + 0.25N(\text{SR}) + 0.15N(\text{Boundary}) + 0.20N(\text{Avg}) + 0.10N(\text{DotBall}) \tag{3}$$

2.2 Bowling Strength Formulation

Similarly, bowling strength combines economy, wicket rate, dot-ball percentage, boundary conceded percentage, and average runs conceded:

$$\text{BowlingStrength} = 0.30N(\text{Economy}) + 0.25N(\text{WicketRate}) + 0.20N(\text{DotBall}) + 0.15N(\text{BoundaryConceded}) + 0.1 \tag{4}$$

2.3 Calculating Match-by-Match Batting and Bowling Performance

To capture how each team performed in every individual match, we aggregated delivery-level data into innings-level statistics and benchmarked them against seasonal averages:

- **Match Batting Performance (Y_i):** Calculated as a team's total innings score divided by the overall seasonal average IPL innings score (≈ 177.16 runs):

$$Y_i = \frac{\text{Team's Innings Score}}{\text{Average IPL Innings Score}} \tag{5}$$

- **Match Bowling Performance (BP_i):** Computed similarly as the inverse relationship of runs conceded relative to the league average scaled by opponent batting strength adjustments:

$$BP_i = \frac{\text{Average IPL Innings Score}}{\text{Runs Conceded in Match}} \tag{6}$$

2.4 Comparator Model Formulation

The regression comparator model combines team strengths, opponent strengths, and home-ground indicators:

$$Y_i = \beta_0 + \beta_1 H_i + \beta_2 BS_i + \beta_3 OBS_i + \epsilon_i \tag{7}$$

Where:

- H_i: Indicator variable (1 if playing at home, 0 otherwise).
- BS_i: Team's batting-strength score.
- OBS_i: Opponent's bowling-strength score.
- ϵ_i: Error term capturing unexplained variation.

3 Variable Results & Numerical Findings

3.1 Final Team Strengths (2022–2026)

The processed historical data yielded the following consolidated strength scores for the four selected teams:

Team Name	Batting Strength	Bowling Strength
Royal Challengers Bengaluru (RCB)	0.9822	0.3791
Mumbai Indians (MI)	0.4816	0.0198
Kolkata Knight Riders (KKR)	0.3364	0.9532
Chennai Super Kings (CSK)	0.2225	0.7122

Table 1: Consolidated Team Batting and Bowling Strength Scores

3.2 Regression Comparator Results

The OLS regression model evaluated on match innings yields the following coefficients for batting performance:

- **Intercept (β_0):** 1.0436 (Standard Error: 0.046, $p < 0.001$)
- **Home Ground Effect (β_1):** -0.0029 ($p = 0.860$)

- **Own Batting Strength (β_2):** 0.0010 ($p = 0.982$)

- **Opponent Bowling Strength (β_3):** -0.0529 ($p = 0.466$)

For bowling performance relative runs conceded:

- **Intercept (β_0):** 1.0422 ($p < 0.001$)

- **Home Ground Effect (β_1):** -0.0232 ($p = 0.156$)

- **Own Bowling Strength (β_2):** -0.0665 ($p = 0.358$)

4 Visualizations & IPython Notebook Integration

In LaTeX, figures and plots generated in Python are typically saved to disk (e.g., as PNG or PDF) and then inserted using the `\includegraphics` command inside a figure environment. Below are the visual representations of the team strength rankings and metric distributions:

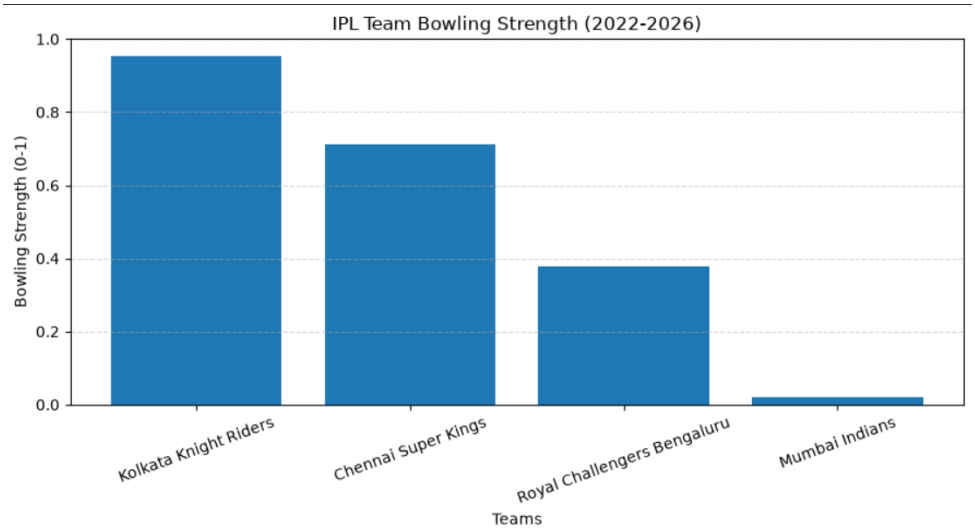


Figure 1: IPL Batting Strength Distribution

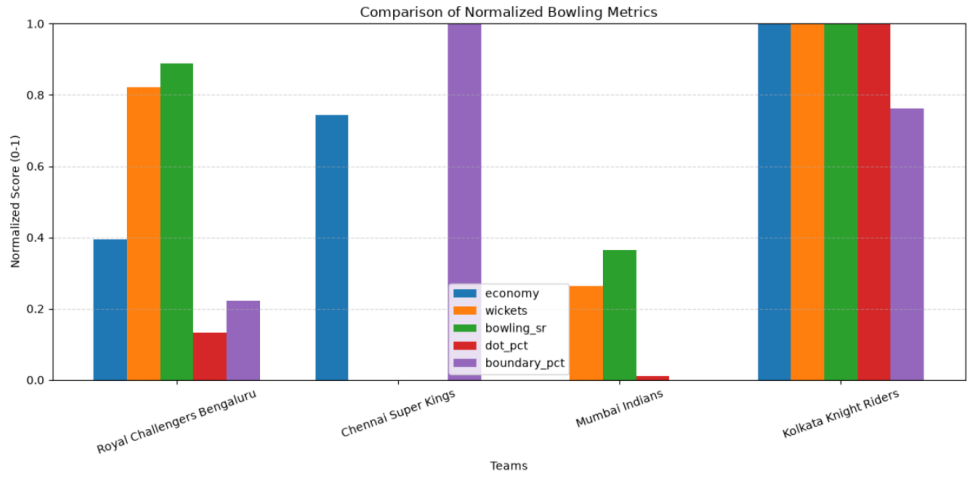


Figure 2: Bowling Metrics Comparison & Heatmap

5 Conclusion

This project successfully synthesized multi-season IPL data to rank MI, RCB, CSK, and KKR. The comparator framework demonstrates that while batting and bowling strengths provide baseline expectations, match dynamics and opponent matchups heavily modulate performance outcomes.